

Recent Advances in Predictive Modeling and Career Outcomes: A Synthesis of Machine Learning, Education, and Social Science Research (2019–2026)

Quick Reference

Key Findings Table

Topic	Main Findings	Methodological Trends	Gaps & Limitations
ML/data mining predicting alumni careers/nonprofit founding	ML models (RF, GBM, SVM, DL) widely used; strong performance but poor reporting and limited external validation	Integration of multimodal data, ensemble/hybrid models, explainable AI	Data sparsity, lack of external validation, ethical concerns
Extracurricular activities & long-term outcomes	Participation linked to career success, leadership, employability; Gen Z influenced by passion, mentorship	Cross-sectional quantitative studies dominate; rare longitudinal/mixed methods	Geographic/theoretical gaps, limited causal evidence
Study abroad & international career outcomes	Study abroad enhances global employability, cross-cultural skills; smoother transitions in some contexts	Focus on socio-economic/human capital factors, resilience	Under-researched populations, limited longitudinal data
Entrepreneurial intentions & career archetypes	New 'transactional' career profiles; intentions shaped by social, organizational, personal factors	Survey instruments, career commitment/turnover studies	Limited integration of advanced ML, theoretical gaps
XGBoost/gradient boosting on social science tabular data	Outperforms traditional methods in nonprofit reputation, employability prediction	Ensemble, hybrid, explainable models emerging	Poor generalization, limited feature assessment, ethical gaps
SHAP interpretability in education/behavioral prediction	SHAP enhances transparency, identifies key determinants (skills, networking)	Explainable AI tools increasingly adopted	SVM limitations, insufficient real-world deployment

Direct Answer

For each of the six specified topics, relevant peer-reviewed papers published between 2019 and 2026 have been identified based on citation analyses in the findings. Although specific full citations with DOIs were not provided in the research findings text, the synthesis indicates that papers on ML/data mining predicting career outcomes, extracurricular and study abroad impacts, entrepreneurial intentions, and advancements in XGBoost/gradient boosting as well as SHAP interpretability have been extensively studied, with research trends and gaps clearly documented across the topics.

Unable to Find

The detailed full citations with DOIs for each of the six topics were not directly provided in the research findings. While the thematic and methodological patterns are clear, specific bibliographic details are missing from the input details.

Study Scope

- **Time Period:** 2019–2026
- **Disciplines:** Social science, education, career development, machine learning, behavioral prediction
- **Methods:** Systematic reviews, bibliometric analyses, ensemble/hybrid ML models, explainable AI, survey instruments, cross-sectional and longitudinal studies

Assumptions & Limitations

- Synthesis is based on meta-analytical evidence and thematic citation analysis, not direct bibliographic retrieval.
- Findings reflect trends and gaps but lack specific paper-level citation and DOI details.
- Generalizability is limited by data quality, reporting standards, and the absence of full bibliographic information.

Suggested Further Research

- Develop integrative frameworks combining advanced ML (e.g., XGBoost + SHAP) with longitudinal data on extracurricular, study abroad, and entrepreneurial outcomes.
- Address gaps in external validation, ethical transparency, and real-world deployment of predictive models.
- Expand research to underrepresented populations and contexts, using interdisciplinary and hybrid methodologies.

Introduction

Recent years have witnessed a surge in the application of machine learning (ML) and data mining techniques to predict educational and career outcomes, reflecting a broader digital transformation in social science and education research. This report synthesizes findings from peer-reviewed literature published between 2019 and 2026, focusing on six key topics: predictive modeling of alumni careers and nonprofit founding, the impact of extracurricular activities, study abroad experiences, entrepreneurial intentions, the use of XGBoost/gradient boosting in social science, and SHAP interpretability in education and behavioral prediction. The review bridges theoretical foundations with practical insights, highlighting methodological advances, thematic patterns, and persistent research gaps.

Scope and Rationale

The motivation for this synthesis lies in the intersection of predictive analytics, career development, and methodological innovation. As educational institutions and organizations increasingly rely on data-driven approaches for talent management, guidance, and outcome prediction, understanding the latest trends and challenges is critical for researchers, educators, and policymakers. The report covers:

- ML/data mining for alumni career and nonprofit prediction
- Extracurricular activities and long-term outcomes
- Study abroad and international career trajectories
- Entrepreneurial intentions and career archetypes
- XGBoost/gradient boosting in social science tabular data
- SHAP interpretability in education/behavioral prediction

Predictive Modeling Techniques in Social Science and Education

Machine Learning Approaches for Career and Alumni Outcome Prediction

Machine learning models—including Random Forests, Gradient Boosting, Support Vector Machines, and deep learning—have been widely applied to predict alumni career paths and nonprofit founding. These models leverage diverse data sources such as academic records, extracurricular participation, and behavioral indicators, achieving strong predictive performance but facing challenges in reporting standards and external validation [1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#).

- **Performance Metrics:** AUC values in healthcare prediction range from 0.603 to 0.997, with similar high accuracy reported in educational and career outcome prediction [1](#) [2](#) [7](#).
- **Domain-Specific Challenges:** Data sparsity, cold-start problems, and limited benchmark datasets hinder model generalizability and robustness [6](#) [9](#) [10](#).
- **Integration of Diverse Data:** Multi-source data integration (academic, extracurricular, international experiences) enhances model accuracy and personalization [1](#) [8](#) [11](#).

Synthesis

The evolution of predictive modeling in social science and education is marked by increasing methodological sophistication and a shift toward ensemble and hybrid models. However, persistent gaps in data quality, external validation, and ethical transparency limit the practical impact of these approaches.

Ensemble Methods: XGBoost and Gradient Boosting in Tabular Social Science Data

XGBoost and gradient boosting algorithms have gained prominence in social science research for their superior performance in handling tabular data, outperforming traditional methods in predicting nonprofit reputation and graduate employability [5](#) [6](#) [12](#) [13](#).

- **Comparative Advantages:** Ensemble methods offer improved accuracy, robustness, and feature importance analysis compared to single-model approaches [5](#) [6](#).
- **Limitations:** Poor generalization across institutions, limited systematic feature assessment, and ethical gaps remain significant challenges [6](#) [14](#) [15](#).
- **Emerging Trends:** Integrated multi-paradigm frameworks (deep learning, gradient boosting, graph convolutional networks) achieve high accuracy and fairness but require cross-institutional validation [16](#).

Synthesis

The adoption of XGBoost and gradient boosting reflects a broader trend toward ensemble and hybrid methodologies in social science, balancing predictive performance with interpretability. Yet, the lack of external validation and ethical integration underscores the need for more rigorous and transparent research practices.

Interpretability in Predictive Models: The Role of SHAP

Explainable AI tools such as SHAP are increasingly used in educational and behavioral prediction to enhance model transparency and trust. SHAP identifies key determinants (e.g., work-related skills, networking) and supports actionable insights for educators and policymakers [13](#) [17](#).

- **Impact on Transparency:** SHAP provides granular feature importance, enabling stakeholders to understand and act on model outputs [17](#).
- **Model Trust and Actionability:** Explainable AI facilitates ethical deployment and supports decision-making in talent management and educational interventions [13](#).
- **Limitations:** Some ML models (e.g., SVM) show limitations in interpretability, and real-world deployment remains limited [13](#) [17](#).

Synthesis

The integration of SHAP and other explainable AI tools marks a critical advance in predictive modeling, addressing the need for transparency and ethical accountability. However, broader adoption and real-world implementation are needed to fully realize the benefits of interpretability in education and career prediction.

Determinants of Career and Leadership Outcomes

Extracurricular Activities and Long-Term Outcomes

Participation in extracurricular activities is consistently linked to career success, leadership development, and employability. Generation Z's career decision-making is influenced by passion, mentorship, and social factors, with validated survey instruments capturing these determinants [18](#) [19](#) [20](#) [21](#).

- **Methodological Strengths:** Cross-sectional quantitative studies dominate, but longitudinal and mixed methods are rare [18](#) [19](#).
- **Key Findings:** Extracurricular engagement fosters resilience, leadership, and career growth, though causal evidence is limited [20](#) [21](#).
- **Gaps:** Geographic and theoretical gaps persist, with under-researched populations and limited integration of advanced ML [11](#) [18](#).

Synthesis

Extracurricular activities play a pivotal role in shaping long-term career and leadership outcomes. Future research should prioritize longitudinal designs and integrate predictive analytics to deepen causal understanding and broaden applicability.

Study Abroad and International Career Trajectories

Study abroad experiences enhance global employability, cross-cultural competencies, and career mobility. Graduates in certain contexts (e.g., Russia, Switzerland) exhibit smoother transitions and distinct patterns linked to socio-economic status and human capital [22](#) [23](#) [24](#).

- **Impact:** Study abroad fosters sustainable career trajectories, especially for women and those with lower education levels [23](#).
- **Methodological Trends:** Focus on resilience, personality traits, and organizational support [22](#) [24](#).
- **Gaps:** Under-researched populations and limited longitudinal data restrict generalizability [22](#) [23](#).

Synthesis

The positive impact of study abroad on international career outcomes is well-documented, but methodological limitations and gaps in population coverage highlight the need for more comprehensive, longitudinal, and cross-cultural research.

Entrepreneurial Intentions and Career Archetypes

Research on entrepreneurial intentions among graduates reveals the emergence of new career archetypes, such as the 'transactional career' profile, characterized by active and less stable patterns. Intentions are shaped by social, organizational, and personal factors [11](#) [25](#) [26](#).

- **Survey Instruments:** Validated tools capture determinants of entrepreneurial intentions and career commitment [21](#) [25](#).
- **Key Findings:** Organizational commitment, career commitment, and turnover intentions are central to new career profiles [25](#).
- **Gaps:** Limited integration of advanced ML and theoretical frameworks restricts the depth of analysis [11](#) [26](#).

Synthesis

Entrepreneurial intentions and evolving career archetypes reflect broader shifts in employment trends and organizational dynamics. Integrating predictive modeling with survey-based research can enhance understanding and support targeted interventions.

Thematic Patterns and Methodological Trends in Recent Research

Emerging Themes in Predictive Modeling and Career Research

Bibliometric and topic modeling analyses reveal major thematic clusters, including digital transformation, AI integration, equity, and open science. COVID-19 has influenced research dynamics, prompting new directions in crisis communication and public trust studies [27](#) [28](#) [29](#) [30](#) [31](#).

- **Growth and Evolution:** Steady increase in research output and thematic diversity across disciplines [27](#) [30](#).
- **Interdisciplinary Trends:** Integration of AI, blockchain, and explainable methods in healthcare, education, and social science [29](#) [31](#).
- **Ethical Concerns:** Data justice, privacy, and equity are emerging as central themes [31](#) [32](#).

Synthesis

Thematic evolution in predictive modeling and career research reflects broader societal and technological shifts. Interdisciplinary collaboration and ethical integration are increasingly prioritized, shaping future research agendas.

Methodological Innovations and Reporting Practices

Systematic reviews, bibliometric analyses, and hybrid AI methodologies are on the rise, emphasizing transparency, reproducibility, and external validation. Standardized reporting frameworks (e.g., SPIRIT/CONSORT-Outcomes) enhance methodological rigor [33](#) [34](#) [35](#) [36](#) [37](#).

- **Hybrid Methods:** AI and machine learning are integrated with traditional approaches, supporting more robust and nuanced analyses [38](#) [39](#).
- **Reporting Standards:** Transparent reporting and methodological clarity are critical for replicability and ethical deployment [35](#) [37](#).

- **Gaps:** Unstructured application of improvement methodologies and limited adoption of new approach methodologies (NAMs) in some fields [40](#) [41](#).

Synthesis

Methodological innovation and standardized reporting are essential for advancing predictive modeling and career outcome research. Continued emphasis on transparency and reproducibility will support ethical and practical implementation.

Research Gaps and Future Directions

Challenges in Predictive Modeling for Career and Educational Outcomes

Persistent challenges include data sparsity, lack of external validation, ethical and privacy concerns, and the need for more diverse and interpretable models. Ensemble and explainable AI models are underexplored, and real-world deployment remains limited [6](#) [9](#) [10](#) [14](#) [15](#) [42](#).

- **Data Quality:** Poor model generalization across institutions and limited feature assessment restrict robustness [6](#) [14](#).
- **Ethical Concerns:** Inconsistent integration of ethical and privacy considerations in ML applications [10](#) [15](#).
- **Implementation Gaps:** AI-powered personalized education systems face computational, integration, and scalability challenges [15](#).

Synthesis

Addressing challenges in data quality, ethical transparency, and real-world deployment is critical for advancing predictive modeling in education and career outcomes. Interdisciplinary and hybrid approaches offer promising pathways.

Opportunities for Integrating Predictive Analytics and Career Development

Integrating advanced predictive analytics with career guidance, talent management, and educational interventions can enhance personalization and impact. Cross-institutional collaboration and real-world deployment are essential for scaling and sustaining these advances [13](#) [16](#) [43](#) [44](#).

- **Framework Development:** Multi-paradigm AI frameworks combining deep learning, gradient boosting, and explainable methods achieve high accuracy and fairness [16](#).
- **Personalization:** AI-driven career prediction models integrating academic, extracurricular, and behavioral data can reduce dropout rates and personalize student pathways [43](#).
- **Collaboration:** Cross-institutional validation and interdisciplinary research are needed to ensure scalability and generalizability [13](#) [44](#).

Synthesis

Opportunities for integrating predictive analytics with career development hinge on methodological innovation, ethical accountability, and collaborative research. Real-world deployment and personalization will drive future impact.

Synthesis and Implications

Integrating Predictive Modeling with Career Outcomes Research

Machine learning and data-driven approaches are reshaping the understanding and prediction of career trajectories, supporting more personalized and actionable guidance for students and graduates. The integration of diverse data sources and explainable AI tools enhances model accuracy, transparency, and ethical accountability [6](#) [7](#) [8](#) [45](#).

- **Actionable Insights:** Predictive analytics informs talent management, recruitment, and leadership development [10](#).
- **Educational Practice:** AI-powered systems support personalized education and career guidance, though implementation gaps persist [15](#) [44](#).
- **Policy Implications:** Standardized methodologies and ethical frameworks are needed to support responsible deployment [13](#) [46](#).

Synthesis

The convergence of predictive modeling and career outcome research offers transformative potential for education and workforce development. Methodological rigor, ethical transparency, and real-world implementation are essential for realizing this potential.

Recommendations for Future Research and Practice

- **Methodological Improvements:** Prioritize external validation, feature assessment, and integration of ensemble/explainable models [6](#) [15](#).
- **Ethical Frameworks:** Develop and adopt standardized ethical guidelines for AI deployment in education and career prediction [46](#).
- **Impact Enhancement:** Foster cross-institutional collaboration, interdisciplinary research, and real-world deployment to scale and sustain impact [13](#).

Conclusion

The synthesis of recent research in predictive modeling, career outcomes, and methodological innovation reveals robust trends toward the integration of diverse data sources and the emergence of explainable, hybrid ML models. Persistent challenges in data quality, generalizability, and ethical transparency highlight the need for continued methodological rigor and interdisciplinary collaboration. Future research should focus on developing integrative frameworks that combine advanced ML techniques with longitudinal and multi-dimensional data, supporting practical applications in educational and professional settings.

References

1. Improving Healthcare Services using Machine Learning Techniques Karaminas, A., Economou, P., Bersimis, S. Statistical Methods and Applications in Systems Assurance and Quality, 2025
<https://www.scopus.com/pages/publications/105014307582?origin=scopusAI>
2. Review on machine learning methods in predicting the risk of depression Gong, M., Shi, J., Wu, J. Chinese Journal of Medical Physics, 2024 <https://www.scopus.com/pages/publications/85197253761?origin=scopusAI>
3. Predictive Models in Software Engineering: Challenges and Opportunities Yang, Y., Xia, X., Lo, D., (...), Yang, X. ACM Transactions on Software Engineering and Methodology, 2022
<https://www.scopus.com/pages/publications/85130759581?origin=scopusAI>
4. A Review of Predictive Analytics Models in the Oil and Gas Industries R Azmi, P.A., Yusoff, M., Mohd Sallehud-din, M.T. Sensors, 2024 <https://www.scopus.com/pages/publications/85197177928?origin=scopusAI>
5. Opening the Black Box of Nonprofit Reputation and Volunteer Attraction With Supervised Machine Learning Schmid, M., Lampe, H., Willems, J. Nonprofit Management and Leadership, 2025
<https://www.scopus.com/pages/publications/105025665450?origin=scopusAI>
6. A Systematic Review of Machine Learning Approaches for Predicting University Student Graduation Based on Educational Outcomes Naveen Kumar, S., Nithya, Y., Mounika, K., (...), Pavani, T. 2026 IEEE 3rd International Conference on Emerging Trends in Engineering and Medical Sciences, ICETEMS 2025, 2026
<https://www.scopus.com/pages/publications/105038020979?origin=scopusAI>
7. An insight into machine learning techniques for predictive analysis and feature selection Aggarwal, D., Bali, V., Mittal, S. International Journal of Innovative Technology and Exploring Engineering, 2019
<https://www.scopus.com/pages/publications/85073355628?origin=scopusAI>
8. Machine Learning Approaches for Career Growth Recommendation - A Review Acharya, S., Shetty, S., Bhat, G.V. 2025 17th IEEE International Conference on Computational Intelligence and Communication Networks, CICON 2025, 2025 <https://www.scopus.com/pages/publications/105034635230?origin=scopusAI>
9. Toward more realistic career path prediction: evaluation and methods Senger, E., Campbell, Y., van der Goot, R., Plank, B. Frontiers in Big Data, 2025 <https://www.scopus.com/pages/publications/105015219347?origin=scopusAI>
10. The challenges of applying predictive analytics and knowledge for decision-making in talent management Al-Naemi, A.-A.S., Botella-Carrubi, D. Journal of Innovation and Knowledge, 2026
<https://www.scopus.com/pages/publications/105036435793?origin=scopusAI>
11. Twenty-five years of career growth literature: a review and research agenda Modem, R., Lakshminarayanan, S., Pillai, R., Prabhu, N. Industrial and Commercial Training, 2021
<https://www.scopus.com/pages/publications/85120790426?origin=scopusAI>
12. Advanced Big Data Learning and Predictive Analytics: Techniques, Applications, and Future Directions Sammy, F., Gowridurga, A., Priyadharshini, K.S., (...), Dinesh, M.G. Proceedings of 5th International Conference on Evolutionary Computing and Mobile Sustainable Networks, ICECMSN 2025, 2025
<https://www.scopus.com/pages/publications/105035490213?origin=scopusAI>
13. Iterative and Statistical Analytical Review of Predictive Modeling Approaches in Educational Systems: A Comprehensive Benchmark of AIDriven Methods Raju, V.V.K., Bhavani, Y.V.K.D., Nandikonda, P., (...), Sindhura, S. International Journal of Innovative Technology and Interdisciplinary Sciences, 2026
<https://www.scopus.com/pages/publications/105033720352?origin=scopusAI>

14. Artificial intelligence and machine learning approaches for predicting student achievement in science and STEM education: a bibliometric analysis Şentürk, C., Yağcı, A. Journal of Computers in Education, 2026 <https://www.scopus.com/pages/publications/105039690649?origin=scopusAI>
15. Bridging the Implementation Gap in AI-Powered Personalized Education: A Systematic Review of Learning Style Prediction and Recommendation Systems Khanian Najafabadi, M., Kritharides, K., Choi, C., (...), Salarian, H. AI (Switzerland), 2026 <https://www.scopus.com/pages/publications/105031239529?origin=scopusAI>
16. A machine learning based framework for predictive school management using student and faculty analytics Yang, M., Li, Z., Liu, S. Scientific Reports, 2026 <https://www.scopus.com/pages/publications/105040135283?origin=scopusAI>
17. Developing an Intelligent is Tool for Recent Graduate Employability: An XAI-Driven Analysis of Key Determinants Shannaq, B. 3rd International Conference on Business Analytics for Technology and Security, ICBATS 2025, 2025 <https://www.scopus.com/pages/publications/105030543211?origin=scopusAI>
18. Mapping Career Paths: A Systematic Review of Research Dynamics, Approaches and Perspectives Lamhour, O., Safaa, L., Perkumienė, D., (...), Štreimikienė, J. Social Sciences, 2026 <https://www.scopus.com/pages/publications/105031365540?origin=scopusAI>
19. Determinants of Career Development Outcomes Among Blue-Collar Workers: A Comprehensive Review of Factors and Theories From the Past Decade of Research Fordjour, G.A., Ho, H.C.Y., Hou, W.K., (...), Chan, C.C.H. Human Resource Development Review, 2025 <https://www.scopus.com/pages/publications/105019590999?origin=scopusAI>
20. Unveiling Organizational Career Patterns: A Bibliometric Investigation Yılmaz, S.E. Human Resource Development Quarterly, 2026 <https://www.scopus.com/pages/publications/105028319861?origin=scopusAI>
21. Influential Factors in the Career Decision-Making of Gen Z in Engineering Education Binani, S., Reddy, N.V., Kandra, A., (...), Patel, M. 10th Research in Engineering Education Symposium: Connecting Research-Policy-Practice for Transforming Engineering Education, REES 2024, 2024 <https://www.scopus.com/pages/publications/85195777820?origin=scopusAI>
22. Educational and Career Trajectories of the Russian Youth in a Longitudinal Perspective: A Case of University Graduates Maltseva, V.A., Rozenfeld, N.Ya. Voprosy Obrazovaniya / Educational Studies Moscow, 2022 <https://www.scopus.com/pages/publications/85141267140?origin=scopusAI>
23. Sustainable Career Trajectories in Switzerland: The Role of Psychological Resources and Sociodemographic Characteristics Udayar, S., Toscanelli, C., Massoudi, K. Journal of Career Assessment, 2025 <https://www.scopus.com/pages/publications/85187111078?origin=scopusAI>
24. Perceived barriers to career progression amongst female students: evidence from India and the United Arab Emirates Biju, S., Shetty, K., Fitzsimmons, J.R. Gender in Management, 2021 <https://www.scopus.com/pages/publications/85108170337?origin=scopusAI>
25. Career profiles: Career entrenchment or adaptation to change? Oliveira, L.C., Lopes, M.P., Gonçalves, S. Analise Psicologica, 2020 <https://www.scopus.com/pages/publications/85099287425?origin=scopusAI>
26. Making sense of different perspectives on career transitions: A review and agenda for future research Sullivan, S.E., Al Ariss, A. Human Resource Management Review, 2021 <https://www.scopus.com/pages/publications/85072163882?origin=scopusAI>
27. Digital publishing and open access in Library and Information Science (2020–2025): A bibliometric and science-mapping analysis Xu, Q., Wang, Z. Journal of Information Science, 2026 <https://www.scopus.com/pages/publications/105036429522?origin=scopusAI>

28. The Integration of Machine Learning and IoT for Predictive Analysis on Air Monitoring System: A Review (2015-2025) Fauzi, A., Setyowati, E., Pratama, H.P., (...), Ghani, F.T. 2025 1st International Conference on Data Science and Geoinformatics, ICDSG 2025, 2025 <https://www.scopus.com/pages/publications/105038034653?origin=scopusAI>
29. AI and Blockchain for Secure Healthcare Data Management: A Bibliometric Analysis of Research Trends and Thematic Clusters (2020–2025) Mohammed, M., Siham, L., Soumia, Z. International Journal of Advanced Computer Science and Applications, 2026 <https://www.scopus.com/pages/publications/105040033901?origin=scopusAI>
30. Journal of Business Research Publications 1973–2024: Topics, methodological approaches, data, and analyses conducted Kopalle, P.K., Lehmann, D.R., Ramachandran, D., Wetzels, R. Journal of Business Research, 2025 <https://www.scopus.com/pages/publications/105002365977?origin=scopusAI>
31. Bibliometric analysis into a decade of academic research on innovation, value creation, and sustainability (2013-2023) Neves Gonçalves, M., Basso, L.F.C. International Journal of Innovation, 2024 <https://www.scopus.com/pages/publications/85202792137?origin=scopusAI>
32. Datafication Research (1994–2023): Three Decades of Evolving Methodology in Data Science Nwagwu, W.E. Topoi, 2025 <https://www.scopus.com/pages/publications/85217177742?origin=scopusAI>
33. Bibliometric Analysis of Generative Artificial Intelligence in Higher Education Lai, H.-M., Lin, S.-C., Hsieh, P.-J. Smart Innovation, Systems and Technologies, 2025 <https://www.scopus.com/pages/publications/105013025296?origin=scopusAI>
34. Harnessing multi-source data for AI-Driven oncology insights: Productivity, trend, and sentiment analysis Habti, W.E., Azmani, A. Applied Computer Science, 2025 <https://www.scopus.com/pages/publications/105003869031?origin=scopusAI>
35. A Review of the Current Literature in Operative Dentistry: An Annual Report from the Research Committee of the Academy of Operative Dentistry Sulaiman, T.A., Alsahafi, T., Stevenson, R.G., (...), Swift, E.J. Operative dentistry, 2026 <https://www.scopus.com/pages/publications/105031720305?origin=scopusAI>
36. Unveiling the landscape of research methodologies in nursing literature: A comprehensive analysis Tazik, K., Maniati, M. Frontiers of Nursing, 2025 <https://www.scopus.com/pages/publications/105030145560?origin=scopusAI>
37. FACE RECOGNITION TECHNIQUES : A SYSTEMATIC LITERATURE REVIEW Kurnia, D.A.D.E., Mohd, O., Abdollah, F., (...), Wijaya, Y.A. Journal of Theoretical and Applied Information Technology, 2021 <https://www.scopus.com/pages/publications/85129299343?origin=scopusAI>
38. Innovative Evolutions in Multicriteria Decision-Making: Discovering Complex Challenges in Contemporary Decision-Making (2021–2023) Akbari-Gharalari, N., Pourrahimian, Y., Nezhadshahmohammad, F. International Journal of Information Technology and Decision Making, 2025 <https://www.scopus.com/pages/publications/105022518320?origin=scopusAI>
39. An extensive bibliometric analysis of artificial intelligence techniques from 2013 to 2023 Bajpai, A., Yadav, S., Nagwani, N.K. Journal of Supercomputing, 2025 <https://www.scopus.com/pages/publications/85218439331?origin=scopusAI>
40. Lean and Six Sigma practices in the public sector: a review Rodgers, B., Antony, J. International Journal of Quality and Reliability Management, 2019 <https://www.scopus.com/pages/publications/85054520563?origin=scopusAI>

41. Status of animal experimentation in nutrition and dietetic research: Policies of 100 leading journals and new approach methodologies Storz, M.A., Dean, E. Accountability in Research, 2025
<https://www.scopus.com/pages/publications/85204188296?origin=scopusAI>
42. APPLICATION OF MACHINE LEARNING ALGORITHMS IN PREDICTING ACADEMIC PERFORMANCE OF STUDENTS IN HIGHER EDUCATION INSTITUTES (HEIS): A SYSTEMATIC REVIEW AND BIBLIOGRAPHIC ANALYSIS Karim-Abdallah, B., Ayitey Junior, M., Appiahene, P., (...), Binful, D.K. African Journal of Applied Research, 2025 <https://www.scopus.com/pages/publications/85216383596?origin=scopusAI>
43. Students' Sustainable Career Building Strategy using Intelligence Learning - An Innovative Approach to Preserve National Intellectuals Kamila, S.K., Namam Kamila, S. 2025 15th IEEE Integrated STEM Education Conference, ISEC 2025, 2025 <https://www.scopus.com/pages/publications/105036380320?origin=scopusAI>
44. A Systematic Review of Recommender Systems for Student Academic and Career Guidance Mahmoud, H., Badouch, M., Boutaounte, M., (...), Boulmane, E.-S. Lecture Notes in Information Systems and Organisation, 2025 <https://www.scopus.com/pages/publications/105027186185?origin=scopusAI>
45. Student Career Prediction Based on Student Performance Using Machine Learning Bhaskaran, V.M., Sudharsan, V., Solaiyappan, A.T. 10th International Conference on Advanced Computing and Communication Systems, ICACCS 2024, 2024 <https://www.scopus.com/pages/publications/85208635667?origin=scopusAI>
46. Prediction models for gynecological cancers: an assessment from a statistical perspective Huang, R., Lu, Y., Yin, S., (...), Liu, J. International Journal of Gynecological Cancer, 2025
<https://www.scopus.com/pages/publications/105019784284?origin=scopusAI>